

Multi-server queueing systems with cooperation of the servers

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Abstract In this paper, we consider an N -server queueing model with homogeneous servers in which customers arrive according to a stationary Poisson arrival process. The service times are exponentially distributed. Two new customer's service disciplines assuming simultaneous service of arriving customer by all currently idle servers are discussed. The steady state analysis of the queue length and sojourn time distribution is performed by means of the matrix analytic methods. Numerical examples, which illustrate advantage of introduced disciplines comparing to the classical one, are presented.

Keywords Multi-server queueing system · Cooperation of the servers · Broadcasting

1 Introduction

Queueing theory was originated by works by A.K. Erlang where he has considered the $M/M/N$ and $M/M/N/N$ type multi-server queues as the models of information transmission in telephone networks. However, the overwhelming majority of the books and papers in queueing theory is now devoted to investigation of more simple single server queues. The variety of disciplines of customers admission and service in multi-server queues is much

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more rich comparing to the single server queues while they did not get enough attention yet. In this paper, we introduce and show usefulness of two new customer's service disciplines in multi-server queues which suggest that, in contrast to the classical discipline where one customer is always served by one server, the service of an arbitrary customer is performed by all servers that are idle at the customer arrival epoch, i.e., servers can cooperate in customer processing.

One discipline suggests that the arriving customer who meets several idle servers at its arrival epoch is copied and each replica of the customer is served by one server independently of other copies of the customer. Customer is considered be served by the system at the earliest service completion of a copy of the customer. While the other copies have to be completely served as well.

The second discipline suggests that the arriving customer who meets m , $m = \overline{2, N}$, idle servers at its arrival epoch is split into m equal parts each of which is considered as an independent customer and is served by the corresponding server with the rate m times higher than the nominal rate of the service of an arbitrary customer by a single server. Customer is considered be served by the system at the latest service completion of the parts of this customer.

We will refer to the first discipline as *BC (Broadcasting with Copying)* discipline and to the second one as *BS (Broadcasting with Splitting)* discipline.

BC discipline can be used in many real-life systems (police and ambulance emergency services, broker-dealer operations, etc.) where speed is a critical factor and customers try to get service using different paths or different channels in the hope that one of them get faster than others and a priori one can not tell which one will. In telecommunications, this discipline may be suitable in modeling *IPTV* (IP television) with fast recovering of the ring (IEEE 802.17).

BS discipline is a reasonable approximation to real-world strategies in current software systems, e.g. web servers. A job is split up into several tasks handled by independent threads in a multi-threaded implementation of a multi-tasking system. Another example is modeling of multimedia downloading from remote servers via *IP* network with help of *ReGet* or *FlashGet* software.

Work of *TCP* protocol in Internet can be modeled by means of the queueing system with *BS* discipline. Peer-to-peer networks is another potential object for modeling by means of the *BS* discipline.

The advantage of these new disciplines consists of better utilization of the system capacity. It is clear also that such service disciplines can be very profitable when the servers (channels) are not reliable and processing (transmission) errors occur. Parallelism essentially increases probability of successful delivering and decreases repetitive service of a customer or its shares. Intuitively probable disadvantage of the introduced disciplines, especially *BC* discipline, consists of redundant servers load and increasing the queue length.

However, due to the better use of the servers capacity and randomness of the service time, there is an intuitive hope that the introduced disciplines allow to reduce an average delivering time of a message. The presented below mathematical and numerical analysis show that this reduction actually takes place and can be essential.

2 Mathematical model

We consider a multi-server queueing system in which customers arrive according to a stationary Poisson arrival process with the rate λ , $\lambda > 0$. The service facility consists of N ,

$N \geq 2$, identical servers. We assume that an arriving customer who meets all servers busy is placed into a buffer of an infinite capacity and then it will be picked up from a buffer according to the *FIFO* (First In First Out) discipline. Service time distribution of this customer by one server will be exponential with parameter μ , $\mu > 0$ (nominal service rate).

Classical (*C*) service discipline assumes that if an arriving customer meets m , $m = \overline{1, N}$, idle servers, it chooses an arbitrary server and is processed by this server during the time which is exponentially distributed with the parameter μ .

Here we consider two new service disciplines suggesting that if an arriving customer meets m , $m = \overline{1, N}$, idle servers, all these servers immediately start the service of this customer.

The *BC* discipline suggests that m copies of this customer are produced and each copy occupies one server as independent customer during the time which is exponentially distributed with the parameter μ . The customer, which was copied, is considered as served by the system when the earliest service completion of one of its copy occurs. But despite of this fact all other copies of this customer continue service independently of each other.

The *BS* discipline suggests that this customer is split into m parts and each part occupies one server as independent customer during the time which is exponentially distributed with the parameter $m\mu$. The customer, which was split, is considered as served by the system only when all parts of the customer finish the service in the system.

We are interested in derivation of the steady state distribution of the queue length and the mean sojourn time in the described system under the introduced disciplines and their comparison with the corresponding distribution and time in the case of using the classical service discipline.

3 Steady state distribution of the number of customers in the system

Let i_t be the number of customers in the system at an epoch t , $i_t \geq 0$. Denote the stationary probabilities of this process by

$$\pi_i = \lim_{t \rightarrow \infty} P\{i_t = i\}, \quad i \geq 0.$$

It can be shown that criterion of these limits existence for the model with all disciplines (*C*, *BS*, *BC*) is the same.

Lemma 1 *The steady-state probabilities π_i , $i \geq 0$, exists if and only if condition*

$$\rho = \frac{\lambda}{N\mu} < 1$$

is satisfied.

In what follows we assume that this condition is fulfilled.

Assertion 1 Average number $L^{(C)}$ of customers in the system in the case of classical service discipline is calculated by

$$L^{(C)} = \pi_0 \left[\sum_{k=1}^N \frac{(N\rho)^k}{(k-1)!} + \frac{N^N}{N!} \frac{\rho^{N+1}}{(1-\rho)^2} (N+1-N\rho) \right]$$

where

$$\pi_0 = \left[\sum_{k=0}^{N-1} \frac{(N\rho)^k}{k!} + \frac{(N\rho)^N}{N!} \frac{1}{1-\rho} \right]^{-1}.$$

This result was derived by A.K. Erlang in the beginning of the 20th century.

The *BC* discipline was investigated recently in the framework of the more general *SM/PH/N* queueing model in the paper (Lee et al. 2006). The following assertion stems from (Lee et al. 2006).

Assertion 2 In the case of the *BC* discipline, the stationary distribution π_i , $i \geq 0$, is computed by

$$\pi_0 = \rho(1-\rho)\sigma_N^{-1}, \quad \pi_i = \pi_0\sigma_i, \quad i = \overline{1, N}, \quad \pi_i = \pi_0\sigma_N\rho^{i-N}, \quad i > N,$$

where

$$\sigma_0 = 1, \quad \sigma_i = \prod_{k=1}^i \frac{\lambda + (k-1)\mu}{k\mu}, \quad i = \overline{1, N}.$$

Average number of customers $L^{(BC)}$ in the system is calculated by

$$L^{(BC)} = N\rho + \frac{\rho^2}{1-\rho} + \rho(1-\rho)\sigma_N^{-1} \sum_{i=0}^{N-1} i\sigma_i.$$

Consider now the most complicated case of the *BS* discipline. It is clear from the description of the model that, at each time epoch, every busy server may have its own current service rate which is independent of service rates of the other servers and is equal to $j\mu$, $j = \overline{1, N}$, if the server started the service of currently processed customer together with $j-1$ other servers.

So, the state of a system can be described by the multi-dimensional Markov process

$$\zeta_t = (i_t, \eta_t^{(1)}, \dots, \eta_t^{(N)}), \quad t \geq 0,$$

where i_t is the number of customers in the system, $i_t \geq 0$, $\eta_t^{(j)}$ is the number of servers currently providing the service with intensity $j\mu$, $j = \overline{1, N}$, at epoch t .

The state space $\mathcal{K}(i_t)$ of the components $\eta_t^{(1)}, \dots, \eta_t^{(N)}$ of the process ζ_t , $t \geq 0$, is defined by formulae

$$\eta_t^{(j)} = 0, \min\{i_t, N\} - \sum_{l=1}^{j-1} \eta_t^{(l)}, \quad j = \overline{1, N-1}, \quad \eta_t^{(N)} = \min\{i_t, N\} - \sum_{l=1}^{N-1} \eta_t^{(l)}.$$

Denote the stationary probabilities of the process ζ_t , $t \geq 0$, by

$$\pi(i, k_1, \dots, k_N) = \lim_{t \rightarrow \infty} P\{i_t = i, \eta_t^{(1)} = k_1, \dots, \eta_t^{(N)} = k_N\}$$

for $\{k_1, \dots, k_N\} \in \mathcal{K}(i)$, $i \geq 0$.

Lemma 2 The stationary probabilities $\pi(i, k_1, \dots, k_N)$ of the Markov chain $\zeta_t, t \geq 0$, satisfy the following system of linear algebraic equations:

$$\lambda \pi(\underbrace{0, \dots, 0}_{N+1}) = \sum_{r=1}^N r \mu \pi(\underbrace{1, 0, \dots, 0}_{r-1}, 1, 0, \dots, 0), \quad (1)$$

$$\begin{aligned} & \left(\lambda + \sum_{j=1}^N j k_j \mu \right) \pi(i, k_1, \dots, k_N) \\ &= \sum_{r=1}^N (k_r + 1) r \mu \pi(i + 1, k_1, \dots, k_{r-1}, k_r + 1, k_{r+1}, \dots, k_N), \quad 0 < i < N, \end{aligned} \quad (2)$$

$$\begin{aligned} & \left(\lambda + \sum_{j=1}^N j k_j \mu \right) \pi(N, k_1, \dots, k_N) = k_1 \mu \pi(N + 1, k_1, \dots, k_N) \\ & + \sum_{r=2}^N (k_r + 1) r \mu \pi(N + 1, k_1 - 1, k_2, \dots, k_{r-1}, k_r + 1, \dots, k_N) \\ & + \lambda \sum_{j=0, k_{N-j} \geq N-j}^{N-1} \pi(j, k_1, \dots, k_{N-j-1}, k_{N-j} - (N - j), \dots, k_N), \end{aligned} \quad (3)$$

$$\begin{aligned} & \left(\lambda + \sum_{j=1}^N j k_j \mu \right) \pi(i, k_1, \dots, k_N) = k_1 \mu \pi(i + 1, k_1, \dots, k_N) \\ & + \sum_{r=2}^N (k_r + 1) r \mu \pi(i + 1, k_1 - 1, k_2, \dots, k_{r-1}, k_r + 1, \dots, k_N) \\ & + \lambda \pi(i - 1, k_1, \dots, k_N), \quad i > N. \end{aligned} \quad (4)$$

Left hand side of (1)–(4) represents the product of the stationary probability of the state of the Markov chain $\zeta_t, t \geq 0$, by the intensity of leaving this state. In turn, the right hand side of (1)–(4) represents the sum of products of the stationary probability of the state of the Markov chain $\zeta_t, t \geq 0$, from which the corresponding state can be reached, by the intensity of suitable transition.

Let us enumerate the states of the components $\{\eta_t^{(1)}, \dots, \eta_t^{(N)}\}$ of the Markov chain $\zeta_t, t \geq 0$, in some order (e.g., the reverse lexicographic order) and form the row-vectors π_i of the stationary-state probabilities $\pi(i, k_1, \dots, k_N)$, corresponding to the state i of the first component of the chain, $i \geq 0$.

Note that the dimensionality of these vectors is equal to the number $K_i = \binom{N+i-1}{i}$ of elements in the set $\mathcal{K}(i), i = \overline{0, N}$, and $K_i = K_N$ for all $i \geq N$.

Let Q be the generator of the Markov chain $\zeta_t, t \geq 0$, with blocks $Q_{i,j}$ corresponding to the transitions of the component $i_t, t \geq 0$, from the state i into the state $j, i, j \geq 0$. The block $Q_{i,j}$ has dimension $K_i \times K_j$ and its entries define intensity of the Markov chain transitions from the state $(i, k_1, \dots, k_N), (k_1, \dots, k_N) \in \mathcal{K}(i)$, into the state $(j, k'_1, \dots, k'_N), (k'_1, \dots, k'_N) \in \mathcal{K}(j), i, j \geq 0, i \neq j$. The non-diagonal entries of the matrix $Q_{i,i}$ have the

same meaning while the diagonal entries are negative and the modulus of the entry defines the total intensity of leaving the corresponding state of the Markov chain.

Lemma 3 *Non-zero entries of the non-zero blocks $Q_{i,j}$ of the generator Q are computed as follows:*

- The diagonal entry of the square matrix $Q_{i,i}$ corresponding to the state $(i, k_1, \dots, k_N)$ is equal to $-(\lambda + \sum_{l=1}^N l k_l \mu)$, $i = \overline{0, N}$.
- The square matrix $Q_{i,i}$ is equal to the matrix $Q_{N,N}$ for all $i, i > N$.
- The entries of the matrix $Q_{i+1,i}$, which are located in the column corresponding to the state $(i, k_1, \dots, k_N)$ and the row corresponding to the state $(i+1, k_1, \dots, k_{r-1}, k_r+1, k_{r+1}, \dots, k_N)$, are equal to $(k_r+1)r\mu$, $i = \overline{0, N-1}$.
- The entries of the matrix $Q_{N+1,N}$, which are located in the column corresponding to the state $(N, k_1, \dots, k_N)$ and the row corresponding to the state $(N+1, k_1-1, \dots, k_{r-1}, k_r+1, k_{r+1}, \dots, k_N)$, are equal to $(k_r+1)r\mu$. The entry located in the column corresponding to the state $(N, k_1, \dots, k_N)$ and the row corresponding to the state $(N+1, k_1, \dots, k_N)$ is equal to $k_1\mu$.
- The square matrices $Q_{i+1,i}$ are equal to the matrix $Q_{N+1,N}$ for all $i, i > N$.
- The entries of the matrix $Q_{i,N}$, which are located in the column corresponding to the state $(N, k_1, \dots, k_N)$ and the row corresponding to the state $(i, k_1, \dots, k_{N-i-1}, k_{N-i} - (N-i), k_{N-i+1}, \dots, k_N)$, are equal to λ , $k_{N-i} \geq N-i$, $i = \overline{0, N-1}$.
- The square matrices $Q_{i,i+1}$ are equal to the matrix λI for all $i, i \geq N$.

Corollary 1 *The structure of the generator Q is given by*

$$Q = \begin{pmatrix} Q_{0,0} & O & O & \cdots & O & Q_{0,N} & O & O & \cdots \\ Q_{1,0} & Q_{1,1} & O & \cdots & O & Q_{1,N} & O & O & \cdots \\ O & Q_{2,1} & Q_{2,2} & \cdots & O & Q_{2,N} & O & O & \cdots \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \ddots \\ O & O & O & \cdots & Q_{N-1,N-1} & Q_{N-1,N} & O & O & \cdots \\ O & O & O & \cdots & Q_{N,N-1} & Q_1 & Q_2 & O & \cdots \\ O & O & O & \cdots & O & Q_0 & Q_1 & Q_2 & \cdots \\ O & O & O & \cdots & O & O & Q_0 & Q_1 & \cdots \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \quad (5)$$

where $Q_0 = Q_{i+1,i}$, $Q_1 = Q_{i,i}$, $Q_2 = Q_{i,i+1}$, $i \geq N$.

Proof of Lemma 2 and Corollary 1 consists of rewriting equilibrium (1)–(4) into the matrix form.

Theorem 1 *Stationary probability vectors π_i , $i \geq 0$, are computed by:*

$$\pi_i = \pi_N B_i, \quad i = \overline{0, N-1}, \quad \pi_N = \pi_{N+1} \Psi, \quad (6)$$

$$\pi_i = \pi_{N+1} \mathcal{R}^{i-N-1}, \quad i \geq N+1, \quad (7)$$

where

$$B_i = \mathcal{M}_{N-1} \cdots \mathcal{M}_i, \quad i = \overline{0, N-1},$$

$$\mathcal{M}_i = -Q_{i+1,i} Q_{i,i}^{-1}, \quad i = \overline{0, N-1}, \quad \Psi = Q_0 \mathcal{A}_N^{-1},$$

$$\mathcal{A}_N = -\left(Q_1 + \sum_{l=0}^{N-1} \mathcal{B}_l Q_{l,N}\right),$$

the matrix $\mathcal{R}$ is the minimal non-negative solution to the system

$$\mathcal{R}^2 Q_0 + \mathcal{R} Q_1 + \lambda I = O,$$

and the vector π_{N+1} is the unique solution to the system of linear algebraic equations

$$\pi_{N+1} [Q_1 + \mathcal{R} Q_0 + \Psi \lambda I] = \mathbf{0}, \quad (8)$$

$$\pi_{N+1} \left[(I - \mathcal{R})^{-1} \mathbf{e} + \Psi \left(\sum_{i=0}^{N-1} \mathcal{B}_i \mathbf{e} + \mathbf{e} \right) \right] = 1.$$

Here I denotes an identity matrix, O denotes zero matrix, $\mathbf{0}$ denotes row vector consisting of 0's, $\mathbf{e}$ denotes column vector consisting of 1's. Dimension of the matrices and vectors should be clear from context.

Analysis of the structure (5) of the generator Q shows that the Markov chain $\zeta_t, t \geq 0$, can be classified as the Quasi-Birth-and-Death process with complex boundary behavior. So, formulae (7) can be derived basing on results by M. Neuts in (Neuts 1981), see also the book (Latouche and Ramaswami 1999). Formulae (6) are derived basing on boundary set of Chapman-Kolmogorov equations for probability vectors $\pi_i, i \geq 0$. Nice structure of the boundary equations (each equation contains only two unknown vectors) allows to get simple relations (6). Equation (8) for the vector π_{N+1} stems from the last boundary equation.

Corollary 2 In the case of the BS service discipline, average number $L^{(BS)}$ of customers in the system is computed by

$$L^{(BS)} = \pi_{N+1} \left[\Psi \left(\sum_{i=1}^{N-1} i \mathcal{B}_i \mathbf{e} + N \mathbf{e} \right) + ((N+1)I - N\mathcal{R})(I - \mathcal{R})^{-2} \mathbf{e} \right]. \quad (9)$$

4 Mean sojourn time

The mean sojourn time $V^{(C)}$ (response time) in the system in the case of the classical service discipline is calculated by

$$V^{(C)} = \frac{1}{\mu} + \left(1 - \pi_0 \sum_{k=0}^{N-1} \frac{(N\rho)^k}{k!} \right) \frac{1}{N\mu - \lambda}. \quad (10)$$

Assertion 3 Mean sojourn time $V^{(BC)}$ of a customer in the system in the case of the BC service discipline is calculated by

$$V^{(BC)} = \rho(1 - \rho) \sigma_N^{-1} \sum_{i=0}^{N-1} \frac{\sigma_i}{(N-i)\mu} + \frac{\rho}{N\mu} \left(N + \frac{1}{1 - \rho} \right). \quad (11)$$

Proof We assume that the sojourn time of the customer in the case of the *BC* service discipline is finished when the first its copy is served. So, if $i, i < N$, customers present in the system at arrival epoch of the tagged customer, then its conditional sojourn time distribution is exponential with the parameter $(N - i)\mu$. This follows from the evident fact that minimum of $N - i$ exponentially distributed with parameter μ random variables is random having exponential distribution with parameter $\mu(N - i)$. If $i, i \geq N$, customers present in the system at arrival epoch of the tagged customer, its sojourn time distribution is defined as convolution of the exponential with the parameter μ distribution of the service time and distribution of the waiting time. The waiting time has Erlangian distribution of order $i - N + 1$ with the parameter $N\mu$. Taking into account formulae for probabilities given in Assertion 2, we conclude proof of Assertion 3. $\square$

Now let us consider the *BS* service discipline. We assumed above that in the case of the this discipline the service time of the customer in the system is the time since the service beginning epoch till the epoch when all parts of the customer shared among servers are served. So, if the customer is served by m servers then the service time ξ_k of the customer in the k th server, $k = \overline{1, m}$, is exponentially distributed with the parameter $m\mu$ and the service time $\xi^{(m)}$ of this customer in the system is defined by

$$\xi^{(m)} = \max_{k=\overline{1, m}} \xi_k.$$

Lemma 4 The random $\xi^{(m)}$ has distribution function defined by formula

$$P\{\xi^{(m)} < x\} = \sum_{k=0}^m \binom{m}{k} (-1)^k e^{-m\mu kx}, \quad x \geq 0.$$

Laplace-Stieltjes transform $\gamma_m(u)$ of this distribution function is computed by

$$\gamma_m(u) = \int_0^\infty e^{-ux} dP\{\xi^{(m)} < x\} = \sum_{k=1}^m \binom{m}{k} (-1)^{k+1} \frac{m\mu k}{m\mu k + u}. \quad (12)$$

Mean value $b_1^{(m)} = E\xi^{(m)}$ of the random $\xi^{(m)}$ is calculated by formula

$$b_1^{(m)} = \sum_{k=1}^m \binom{m}{k} (-1)^{k+1} \frac{1}{m\mu k}.$$

Let $W(x)$ and $V(x)$ be distribution functions of waiting and sojourn time of an arbitrary customer in the system and $v(u)$ and $u(u)$ be their Laplace-Stieltjes transforms:

$$w(u) = \int_0^\infty e^{-ux} dW(x), \quad v(u) = \int_0^\infty e^{-ux} dV(x), \quad \operatorname{Re} u > 0.$$

Theorem 2 Laplace-Stieltjes transform $v(u)$ of the sojourn time distribution is calculated by

$$v(u) = \sum_{i=0}^{N-1} \pi_i \gamma_{N-i}(u) \mathbf{e} + [\pi_N + \pi_{N+1} (I - \mathcal{R}\mathcal{Z}(u))^{-1} \mathcal{Z}(u)] \tilde{\mathcal{Z}}(u) \mathbf{e} \gamma_1(u), \quad (13)$$

where

$$\mathcal{Z}(u) = (uI - Q_1 - Q_2)^{-1} Q_0, \quad \tilde{\mathcal{Z}}(u) = (uI - Q_1 - Q_2)^{-1} Q_{N,N-1},$$

functions $\gamma_m(u)$, $m = \overline{1, N}$, are defined by formula (12).

Proof follows from the formula of the total probability and the following two observations. If the tagged customer meets i , $0 \leq i < N$, busy servers, it does not wait in a queue and it is served by $(N - i)$ servers and so the Laplace-Stieltjes transform of the sojourn time distribution is defined by the function $\gamma_{N-i}(u)$. If the tagged customer meets all servers busy, it will wait in a queue and then it will be served by only one server and so the Laplace-Stieltjes transform of the sojourn time distribution is the product of Laplace-Stieltjes transforms of the waiting and service time distributions.

Derivation of the formula

$$w(u) = \sum_{i=0}^{N-1} \pi_i + \pi_N \tilde{\mathcal{Z}}(u) \mathbf{e} + \pi_{N+1} [I - \mathcal{R}\mathcal{Z}(u)]^{-1} \mathcal{Z}(u) \tilde{\mathcal{Z}}(u) \mathbf{e} \quad (14)$$

for the Laplace-Stieltjes transform $w(u)$ of the waiting time distribution is implemented by means of the method of collective marks (method of catastrophes), see, e.g. van Danzig (1995), Kasten and Runnenburg (1956). We interpret the parameter u as an intensity of some imaginary stationary Poisson flow of catastrophes. So, $w(u)$ has meaning of probability that catastrophes do not arrive during the waiting time of an arbitrary customer.

It is obvious that waiting time of the arbitrary (tagged) customer does not depend on the arrival process after the tagged customer arrival epoch. Because the waiting time is equal to zero if the customer meets free servers and so probability that catastrophes do not arrive during the waiting time is equal to 1, the first summand in the right side of formula (14) is clear. Now consider situation when the tagged customer meets all servers be busy.

Denote $w(u, i, k_1, \dots, k_N)$ probability of no one catastrophe arrival during the waiting time of the tagged customer conditional it meets i , $i \geq 0$, customers in the queue and the state $(k_1, \dots, k_N)$ of the processes $\eta_t^{(1)}, \dots, \eta_t^{(N)}$ which define the number of servers currently having service rate $\mu, 2\mu, \dots, N\mu$, correspondingly. Analyzing the behavior of the system until the first transition of the Markov process ζ_t , $t \geq 0$, after the tagged customer arrival epoch and using the formula of total probability we get the difference equations

$$\begin{aligned} w(u, i, k_1, \dots, k_N) &= \left[\frac{\sum_{(k'_1, \dots, k'_N) \in \mathcal{K}(N), (k'_1, \dots, k'_N) \neq (k_1, \dots, k_N)} (Q_1)_{(k_1, \dots, k_N), (k'_1, \dots, k'_N)}}{u - (Q_1)_{(k_1, \dots, k_N), (k_1, \dots, k_N)}} \right. \\ &\quad + \frac{\sum_{(k'_1, \dots, k'_N) \in \mathcal{K}(N)} (Q_2)_{(k_1, \dots, k_N), (k'_1, \dots, k'_N)}}{u - (Q_1)_{(k_1, \dots, k_N), (k_1, \dots, k_N)}} \Big] w(u, i, k'_1, \dots, k'_N) \\ &\quad + \frac{\sum_{(k'_1, \dots, k'_N) \in \mathcal{K}(N)} (Q_0)_{(k_1, \dots, k_N), (k'_1, \dots, k'_N)}}{u - (Q_1)_{(k_1, \dots, k_N), (k_1, \dots, k_N)}} w(u, i-1, k'_1, \dots, k'_N), \\ i &\geq 1, \quad (k_1, \dots, k_N) \in \mathcal{K}(N). \end{aligned} \quad (15)$$

Here $(Q_{j,j'})_{(k_1, \dots, k_N), (k'_1, \dots, k'_N)}$ denotes the entry in the row corresponding to the state $(k_1, \dots, k_N)$ and column corresponding to the state $(k'_1, \dots, k'_N)$ of the matrix $Q_{j,j'}$.

The ratios in (15) define probability of a possible transition of the processes $(\eta_t^{(1)}, \dots, \eta_t^{(N)})$, and no catastrophe arrival before this transition. The corresponding multipliers have the meaning of a probability of no catastrophe arrival after this transition until the end of the waiting time of a tagged customer.

Let us introduce the column vectors $\mathbf{w}(u, i)$ of the Laplace-Stieltjes transforms $w(u, i, k_1, \dots, k_N)$ listed in the reverse lexicographic order, $i \geq 0$. Rewriting (15) in the matrix form, we have relation

$$(uI - Q_1 - Q_2)\mathbf{w}(u, i) = Q_0\mathbf{w}(u, i-1), \quad i \geq 1.$$

By introducing denotation $\mathcal{Z}(u)$, we rewrite these equations as

$$\mathbf{w}(u, i) = \mathcal{Z}(u)\mathbf{w}(u, i-1), \quad i \geq 1. \quad (16)$$

In the same way, we derive equation

$$\mathbf{w}(u, 0) = \tilde{\mathcal{Z}}(u)\mathbf{e}. \quad (17)$$

Relations (16) and (17) yield

$$\mathbf{w}(u, i) = (\mathcal{Z}(u))^i \tilde{\mathcal{Z}}(u)\mathbf{e}, \quad i \geq 0. \quad (18)$$

Now, formula (14) directly follows from the formula of total probability and formulae (7), (18). Theorem is proved. $\square$

Corollary 3 *The mean sojourn time $V^{(BS)}$ of an arbitrary customer in the system in the case of the BS discipline is computed by*

$$\begin{aligned} V^{(BS)} = & \sum_{i=0}^{N-1} \pi_i b_1^{(N-i)} \mathbf{e} + \pi_N (\tilde{\mathcal{S}}^{(1)} + \tilde{\mathcal{S}}^{(0)} + Ib_1^{(1)}) \mathbf{e} \\ & + \pi_{N+1} \mathcal{T} [\mathcal{R} \mathcal{S}^{(1)} \mathcal{T} \mathcal{S}^{(0)} \tilde{\mathcal{S}}^{(0)} + \mathcal{S}^{(0)} \tilde{\mathcal{S}}^{(1)} + \mathcal{S}^{(1)} \tilde{\mathcal{S}}^{(0)} + \mathcal{S}^{(0)} \tilde{\mathcal{S}}^{(0)} b_1^{(1)}] \mathbf{e}, \end{aligned}$$

where

$$\begin{aligned} \mathcal{S}^{(0)} &= \mathcal{F} Q_0, & \mathcal{S}^{(1)} &= \mathcal{F} \mathcal{S}^{(0)}, & \tilde{\mathcal{S}}^{(0)} &= \mathcal{F} Q_{N,N-1}, & \tilde{\mathcal{S}}^{(1)} &= \mathcal{F} \tilde{\mathcal{S}}^{(0)}, \\ \mathcal{T} &= (I - \mathcal{R} \mathcal{S}^{(0)})^{-1}, & \mathcal{F} &= (-Q_1 - Q_2)^{-1}. \end{aligned}$$

5 Numerical examples

As it was noted in Introduction, intuitively possible disadvantage of the disciplines introduced in this paper, especially of the BC discipline, consists of increasing the mean queue length. However, it was guessed that the introduced disciplines allow to reduce the customer sojourn time. The aim of the implemented numerical experiment is to show that the introduced disciplines can actually provide essential profit in this respect comparing to the classical discipline.

The following figures present dependence of the two performance measures (average number L of customers in the system and average sojourn time V (response time)) for the

Fig. 1 Dependence of the average number L of customers in the system on the traffic intensity ρ

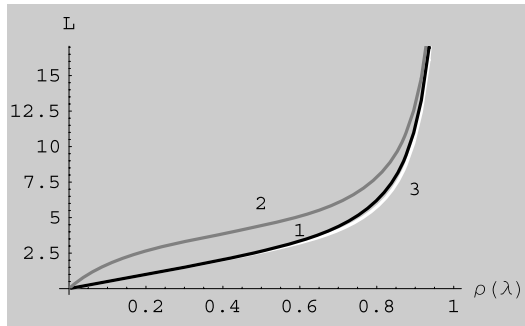
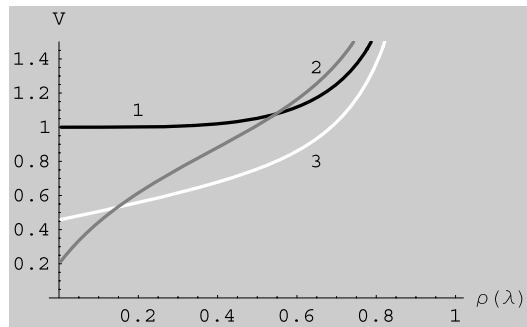


Fig. 2 Dependence of the average sojourn time of customers in the system on the traffic intensity ρ



$M/M/N$ system with the C (classical) service discipline (curve number 1), BC discipline (curve number 2), and BS discipline (curve number 3) on system parameters.

Figures 1 and 2 illustrate dependence of these performance measures on the traffic intensity (load) $\rho = \frac{\lambda}{N\mu}$. The number N of servers is equal to 5. Nominal service intensity μ is equal to 1. The load ρ is varied by means of variation of the intensity λ of arrival process ($\rho = \rho(\lambda)$).

It is worth to note that, if we ignore the reliability aspects, the BC discipline seems to be a bit less reasonable discipline comparing to the BS discipline because it requires redundant work of the servers. Only one, the fastest, copy of a customer actually is required while all the others are just deleted after their service. However, the advantage of the BC discipline consists of more simple implementation: the arriving customer is just sent to all the idle servers, while the BS discipline requires additional work to split a customer into parts and then, after the service, to combine the parts into one entity again. Moreover, as it is evidently seen on Fig. 2, the BC discipline can provide better value of response time comparing to the BS discipline when the input rate is very small. It can be easily explained in our example analytically. When the value of λ is close to zero, a customer usually meets busy servers very rarely, it does not wait and is served by all 5 servers simultaneously. So, its response time is defined by the service time. In the case of the BC discipline, the time until the earliest copy completion is equal to $\frac{1}{5\mu} = 0.2$. In the case of the BS discipline, the time until the service completion of the latest part of a customer is about 0.456666.

Figures 3 and 4 illustrate dependence of the system performance measures on the number of servers N . This number varies from 1 to 6. Service intensity μ is equal to 1. Traffic intensity ρ is equal to 0.8 in the Fig. 3 for L and ρ is equal to 0.2 in the Fig. 4.

Fig. 3 Dependence of the average number L of customers in the system on the number of servers N

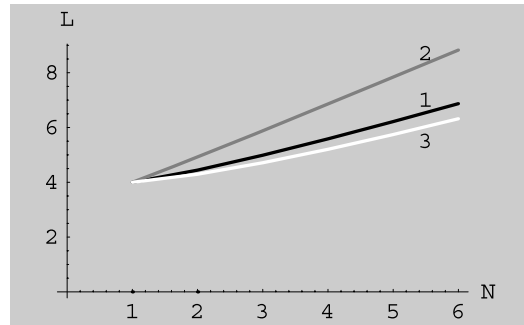
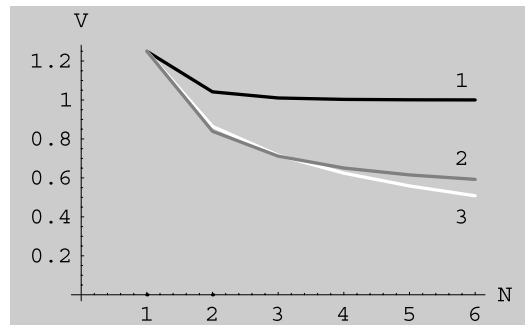


Fig. 4 Dependence of the average sojourn time of customers in the system on the number of servers N



One can see that, under the fixed value of the traffic intensity, the profit provided by using the introduced strategies grows when the number of servers N increases.

As follows from numerical results, the disciplines BS and BC allow to provide essentially better response time comparing to the classical discipline.

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References

- van Danzig, D. (1955). Chaines de Markof dans les ensembles abstraits et applications aux processus avec regions absorbantes et au probleme des boucles. *Annales de l'Institut Henri Poincare*, 14(3), 145–199.
- Kasten, H., Runnenburg, J.T. (1956). *Priority in waiting line problems*. Amsterdam: Mathematisch Centrum.
- Lee, M. H., Dudin, A. N., & Klimenok, V. I. (2006). The $SM/M/N$ queueing system with broadcasting service. *Mathematical Problems in Engineering*, 2006. Article ID 98171, 18 pages.
- Neuts, M. F. (1981). *Matrix-geometric solutions in stochastic models—an algorithmic approach*. Baltimore: Johns Hopkins University Press.
- Latouche, G., & Ramaswami, V. (1999). *Introduction to matrix geometric methods in stochastic modelling, Series in statistics and applied probability*. Philadelphia: ASA-SIAM.